# Carlo Makeever

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## Education

**San Diego State University** | *B.S. Mechanical Engineering* | GPA: 3.71 — Expected May 2028
**San Diego Mesa College** | *A.S. Computer Science & Engineering* | GPA: 3.89 — 2023 – 2026

## Experience

**NASA Jet Propulsion Laboratory** — Pasadena, CA
*Thermal Engineering Intern, MSL Thermal Systems* — *Jun 2025 – Nov 2025*

- Supported Curiosity heater-table updates and thermal-margin assessments for seasonal Martian loads, including an update deployed to the rover.
- Built Python tools to integrate 12+ years of thermal and heating-event data into time-aligned datasets for engineering analysis, operations, and anomaly review.
- Automated hourly heating-event data ingestion and validation (Bash, cron); supported daily downlink analysis and command validation.

**San Diego State University, Materials Laboratory** — San Diego, CA
*Aerospace Materials Research Assistant* — *May 2026 – Present*

- Developing an Ansys APDL workflow for sequential thermal-to-structural FEA of layered ceramic tiles with temperature-dependent properties and convection/radiation boundary conditions.
- Computed effective thermal conductivity and stiffness inputs for carbon/epoxy representative volume elements at 70% fiber volume using Ansys Material Designer.
- Built a Python transfer-matrix solver verified against published results; implemented differential evolution to optimize multilayer coatings for reduced mid-wave infrared signature.

**SDSU, Computational Materials Research Laboratory** — San Diego, CA
*Undergraduate Research Assistant* — *Jul 2026 – Present*

- Evaluated monolayer $MoTe_2$ phase stability using VASP/PBE density functional theory; computed a phase-energy difference of 43 meV per formula unit versus a published value of 40 meV.

## Engineering Projects

**SDSU Rocket Project**
*Lower Airframe Lead, Structures & Materials* — *Aug 2026 – Present*

- Lead a 13-member team developing a liquid rocket lower airframe and fin can; define load cases and acceptance criteria.
- Direct spar trade studies comparing cross-section, material, and isogrid pocketing using hand calculations and structural FEA to evaluate mass-stiffness tradeoffs.
- Lead design reviews of three fin-can concepts; identified a dimensional incompatibility ruling out the legacy spar frame.

**SDSU Aerospace Navigation and Control Society**
*Structures and Materials Lead* — *Aug 2026 – Present*

- Designed, built, and flew a Level 1 high-power rocket using SolidWorks and OpenRocket; documented engineering requirements and verification checks.
- Increased fin thickness from 3.03 to 4 mm for a calculated 2.3x bending stiffness at 1.32x mass.
- Identified fin-tab interference with a centering ring in CAD before fabrication.

**SDSU Experimental Flight Sciences**
*Founder* — *Oct 2026 – Present*

- Modeled a 1.5 m-span UAV testbed in OpenVSP and built a containerized OpenFOAM RANS workflow to evaluate nose shapes and fineness ratios.
- Developed preliminary thrust-drag budgets across four turbine configurations to compare propulsion concepts and assess high-speed RC aircraft feasibility.

## Technical Skills

**Design & Fabrication:** SolidWorks, OpenVSP, OpenRocket, composite layup, 3D printing
**Engineering Analysis:** Thermal/structural FEA, CFD, Ansys (Mechanical, APDL, Material Designer), OpenFOAM
**Programming & Tools:** Python, MATLAB, C/C++, Bash, Git, Linux, high-performance computing